

**SMART POLYPHARMACY & VITAMIN
RECOMMENDATION SYSTEM FOR GERIATRIC
PATIENTS**

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**PERSONALIZED FOOD RECOMMENDATION
SYSTEM FOR VITAMIN DEFICIENCY**

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DECLARATION OF THE CANDIDATE AND SUPERVISOR

I declare that this is my own work and this proposal does not incorporate without acknowledgment any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of my knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgment is made in the text.

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ABSTRACT

Polypharmacy is the long-term use of more than one medication (medicines), and in the growing populations of older adults and people with chronic conditions, it is becoming increasingly common; however, while polypharmacy is essential for properly managing multiple complex health conditions, it also has unintended side effects, including adverse drug-nutrient interactions, a decreased ability to absorb certain nutrients, and ultimately deficiencies in multiple vitamins, which can negatively affect recovery from illness, cause extreme fatigue and result in quality of life for patients to decrease. Typical dietary recommendations to patients focus primarily on generalized nutrition or calorie counting and do not account for the clinical condition of patients, the effects of their medications, or their actual diet-related behaviors.

This study introduces a Smart Polypharmacy and Vitamin Recommendation System, which is an intelligent, patient-centric solution designed to address the gap created by translating the patient's clinical diagnosis into an actual diet-related behavior with respect to clinical dietary recommendations. The system compiles patient-specific data, i.e., age, body mass index (BMI), current medical conditions, medications, and dietary restrictions, to create individual meal plans for patients with vitamin deficiencies. This solution does not just give patients recommendations on what they should eat but continuously monitors if the patient is following the prescribed diet.

This new system contains many advanced features such as the real-time tracking of meals consumed; adherence analysis; artificial intelligence (AI) assisted meal recommendations; and adaptive mechanisms to substitute one food item for another. Gamification includes a method for tracking streaks, awarding points, and visualizing progress toward your goals, while an interactive chat assistant provides personalized dietary assistance. This system will produce clinically structured reports that help healthcare professionals monitor how their patients are progressing and make appropriate decisions.

This system is designed to be a dynamic, adaptive, and measurable healthcare solution by utilizing AI, user behaviour tracking, and the principles of clinical nutrition. In addition, this system will improve compliance with dietary recommendations and increase patient engagement with the healthcare system by creating personalized treatment plans. Ultimately, this system will help to better manage vitamin deficiencies in patients on multiple medications and assist them in living healthier and more fulfilling lives.

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List of abbreviations

Abbreviation	Full Form
AI	Artificial Intelligence
ML	Machine Learning
CDSS	Clinical Decision Support System
BMI	Body Mass Index
mHealth	Mobile Health
EHR	Electronic Health Record
SaaS	Software as a Service
API	Application Programming Interface
UI	User Interface
UX	User Experience
PDF	Portable Document Format
CRUD	Create, Read, Update, Delete
DB	Database
NoSQL	Not Only Structured Query Language
HTTP	Hypertext Transfer Protocol
HTTPS	Hypertext Transfer Protocol Secure
JSON	JavaScript Object Notation
REST	Representational State Transfer
JWT	JSON Web Token
GDPR	General Data Protection Regulation
B12	Vitamin B12
DDI	Drug–Drug Interaction
DNI	Drug–Nutrient Interaction
NLP	Natural Language Processing
HCI	Human–Computer Interaction
SPVRS	Smart Polypharmacy & Vitamin Recommendation System
DR	Dietary Recommendation
DA	Dietary Adherence
ARS	Adaptive Recommendation System

1 Introduction

1.1 Background and Literature Survey

The rising number of older people in our world has caused serious problems for current healthcare systems when it comes to dealing with older patients who have many different chronic illnesses at once (known as "multi-morbidity"). The longer people live, the more of them will experience chronic medical conditions (or chronic illnesses that happen over a long time) such as: diabetes, high blood pressure (hypertension), heart disease/cardiovascular disease, renal disease, etc., and these illnesses usually require continuous medicine/medication use. Because of this, many older patients are prescribed 5 or more (also known as "polypharmacy") different types of medicines regularly. While polypharmacy helps some people manage very complex/challenging diseases well, it can also create risks such as adverse drug reactions, drug-interactions (drugs affecting each other), and more importantly nutritionally-related drug interactions.

Interactions between drugs and food can occur when medications affect how our bodies absorb, metabolize and eliminate nutrients. Drug-nutrient interactions affect the bioavailability of numerous essential vitamins and minerals, which over time can lead to nutritional imbalances or deficiencies. For example: one common drug used to treat diabetes is Metformin, which has been associated with decrease absorption of vitamin B12 due to the effect that it has on the calcium dependent receptor-mediated absorption system in the terminal ileum [2]. Another class of drugs are the diuretics, which can cause the excretion of potassium and magnesium. Proton pump inhibitors may cause there to be a decrease in the serum vitamin B12 due to the reduction in gastric acid secretion [4]. The clinical impact of nutritional deficiencies secondary to drug-nutrient interactions can result in the following clinical symptoms and health issues: megaloblastic anaemia, peripheral neuropathy, impaired cognitive functioning; longer recovery times from illnesses.

Even though drug-nutrient interactions are well documented, there is not a great emphasis placed on nutritional management in the healthcare system. The majority of the time healthcare systems focus mainly on treating the disease process and not the importance of providing preventive nutrition management (food/nutrition); therefore, too many patients are being treated with medication only without any appropriate nutritional support thus leading to multiple long-term complications. This shows us the importance of integrating systems (clinical interventions). These systems must incorporate both clinical and personal nutrition management in order to provide better health outcomes for patients.

Standardized dietary recommendations have been largely based on standardized food guidelines (food pyramids, dietary reference intakes, general food charts) which give us important general information about good nutrition but are generic and therefore, not tailored to the individual. Individual variability is influenced by factors such as age, sex, BMI, medical conditions and medications, lifestyle habits and cultural preferences related to food. Research in personalized nutrition has proven that people who receive individualized dietary interventions based on their individual phenotype, genotype and dietary information show much higher rates of success in terms of dietary changes than people who receive generalized population-based dietary recommendations. The pan-European Food4Me Study showed that people who received personalized advice experienced much greater improvement in dietary behaviors than people who received non-personalized advice [1].

While these limitations pose a challenge, digital health technologies are emerging as viable solutions for improving dietary management. The growing use of mobile health (mHealth) applications, wearable technologies, and online nutrition platforms is attributed to their accessibility and convenience. Through these technologies, individuals can track food consumption on a daily basis, monitor physical activity, and receive feedback on their eating behavior. However, most of these applications have been developed with

general health and wellness in mind, but not specifically for a clinical population. They typically only provide users with information on their daily calorie, macronutrient, and weight/energy level, without taking into consideration important medical variables like disease state(s), medication effects, and potential micronutrient deficiencies. Systematic evaluations of popular nutrition apps have determined that micronutrient intake estimates are typically inconsistent across multiple apps and therefore are not clinically accurate. Additionally, due to the reliance of these technological systems on self-reporting of dietary intake data, the validity of these systems is generally low. Users must self-document their own food and beverage items, which can be very time consuming and lead to inaccuracies; studies have shown that self-reported dietary intake data often contains significant amounts of error due to recall bias, estimation error, misreporting by the user, and/or the lack of motivation for the user to complete the documentation to accurately reflect actual consumption. Therefore, there exists a disparity between documented and actual dietary intake, thereby making the accurate assessment of dietary intake and adherence much more difficult.

New implementations of Machine Learning (ML) and Artificial Intelligence (AI) are evolving how we manage food and health. AI can help analyze large volumes of patient data to discover complex relationships that could not previously be found. This includes using decision trees, support vector machines, and neural networks to identify if a patient has a vitamin deficiency and help them create their own meal recommendations based on their chronic conditions [6]. A review of ML and nutrition in research identified clustering, collaborative filtering, and deep neural networks as effective tools for creating recommendations based on chronic disease profiles [5]. In addition, CDSSs can be of assistance by providing customized, diet recommendation plans based on individual patient parameters. CDSSs are created by combining clinical data, nutritional knowledge, and algorithmic models to provide recommendations for optimal dietary intake. For example, a fuzzy logic-based CDSS has been developed for patients with multiple chronic diseases that is approximately 97% accurate in reproducing the recommendation of a nutritionist [7].

Even though the methods that were previously discussed are all useful, they each have their limitations such as not taking into account how well a person adheres to their diet. New research is also showing that nutrition can be assessed through computer vision techniques, by analyzing images of foods, estimating the size of servings and identifying signs of nutrient deficiencies. For instance, examining the skin, nails and eyes can provide clues to the presence of anaemia, vitamin deficiencies and so on. Although some of these techniques will enable innovative ways to detect and monitor these avian functions, they are still very much in an experimental stage and, therefore, have not been widely implemented in practical health care delivery systems.

Optimization approaches to dietary planning have been used and could provide some innovative ways of creating a meal plan that meets specific nutritional needs using linear programming, genetic algorithms, constraint satisfaction modelling techniques, etc. These techniques can create meal plans with an effective distribution of nutrients but often do not take into consideration factors such as availability, cost, cultural preference or the ability of the individual patient to adhere to the diet. Therefore, the meal plans that are generated by these techniques will likely be unworkable or unacceptable to the user.

The lack of real-time monitoring and feedback mechanisms is one of the major limitations of currently available systems. Most applications provide static recommendations for people but do not provide the ability for the application to continuously monitor user activity and respond to the changing status of the user (e.g., worsening or improving health). Dietary needs change dynamically due to various factors such as changes in medications, progression of disease, and changes in the patient's lifestyle. Thus, there is a need for adaptive systems to continuously monitor these needs and provide real-time recommendations based on the current situation.

Another important factor that is frequently overlooked when considering application development for clinical nutrition is user engagement. Behavioral factors play a major role in whether or not individuals will follow dietary recommendations provided to them. Most currently available systems do not have any motivational characteristics that encourage long-term use of an application. Research has demonstrated that by incorporating game-like elements into mobile health applications (e.g., reward systems, record of progress towards a goal, achievement systems, etc.) patients will have improved engagement with a mobile health application and subsequently adhere to good health practices. A review of gamification and medication adherence within mobile health applications that used gamified features found that most of the studies had improved or maintained the condition of the patient receiving medication compared to those without gamified features (i.e. a few studies showed virtually no difference when comparing gaming and non-gaming applications) [8]. Gamification features are rarely found within clinical nutrition platforms.

There are also limitations of the current system due to inaccessibility and inadequate support which restrict their effectiveness. Patients often need guidance when deciding on what to eat, largely due to not having access to recommended foods, or due to not having recommended foods that meet their dietary requirements. Without intelligent assistants or interactive support systems to assist you with this challenge, it is hard for the user to find a way through the available dietary solution to resolve their challenge.

Another key limitation faced by current systems is the lack of integration between dietary systems and clinical health care environments. Most currently available nutrition applications are used independently; thus, they do not have structured feedback or reporting methods available that would allow for their use by clinical health care providers. This disjoint reduces the overall ability of the systems to have meaning in clinical practice and ultimately restricts how many physicians will adopt them into their practice. There is a need for the clinical health care profession to have reliable data and analytical feedback from nutrition systems to monitor the progress of patients and to make appropriate medical decisions. A scoping literature review of diet-related health recommender systems has found that there are several key gaps between current systems, e.g. user-centered design, actionable guidance, and standardized evaluation processes [9, 10].

Moreover, many of the current systems fail to provide useful analytical information from data collected. Many current systems contain basic data tracking and reporting, however the vast majority do not have advanced analytics, such as adherence trends, patterns of deficiencies, or predictive models. Access to such analytical information is necessary to improve the outcomes of patients and to optimize their treatment programs.

The current research trends are advocating for the development of integrated and intelligent health care systems that offer multiple capabilities within one system. The purpose of these systems is to better connect clinical recommendations and how patients behave in the real world by offering personalization, monitoring, adaptation, and feedback loops. When executed effectively, these systems will leverage AI, data analytics, and user-centered designs in order to provide more effective and practical solutions for managing healthcare.

The Smart Polypharmacy and Vitamin Recommendation System will expand on the current research trends by providing a holistic and adaptive method for nutritional management. Unlike traditional methods, the Smart Polypharmacy and Vitamin Recommendation System produce personalized meal plans based on patient-specific information while simultaneously tracking food intake and evaluating adherence to the designed meal plan. The Smart Polypharmacy and Vitamin Recommendation System includes features such as intelligent food recommendations, gamification-based engagement, intelligent meal substitution, and interactive chat support. The Smart Polypharmacy and Vitamin Recommendation System also produces

clinically structured reports that empower healthcare professionals to evaluate patient progress and determine evidence-based actions.

To summarise, while previously completed research on nutrition technology and healthcare systems have had a significant impact on research in those areas, many solutions currently exist, with a majority being limited in terms of their wide-ranging impact and functionality. They typically focus on an individual aspect, for example; recommendations, tracking, or assessment, and do not contain a comprehensive integrated solution. Therefore, there currently exists a need for a comprehensive system that can integrate personalised recommendations, real world monitoring, adaptive learning, and clinical assistance within the same system. The research presented in this report addresses that systemic need by creating a "smart" patient centered solution that improves the efficacy and overall practicality of dietary management for polypharmacy patients.

1.2 Research Gap

There are many improvements to health care and the area of nutrition science; however, there are still significant barriers to successfully treating vitamin deficiencies in those on multiple medications (polypharmacy). Based on a systematic review of the literature and existing systems, five major areas where more research is needed have been established:

Research Gap A – Lack of Gamification and User Engagement Mechanisms in Clinical Nutrition Systems

There is one aspect of current nutrition management systems that is very important - they do not have any elements that encourage the user to continue using these tools. In other words, current nutrition management systems do not provide user engagement features or motivation to help improve patients' compliance with their prescribed dietary recommendations. This becomes even more troublesome for individuals taking multiple medications, as they often require compliance for long periods of time with their dietary recommendations. Gajardo Sánchez et al. (2023) confirmed that including gamification strategies (points, badges, leaderboards, etc.) as part of the healthcare management system significantly increases the level of engagement and commitment by patients to follow their prescribed dietary recommendations, as well as providing a means for patients to take a more active role in managing their chronic medical conditions [11].

In a similar manner to Dehghani et al. (2023), Gamification techniques can improve motivation and adherence to the treatment of chronic diseases in mobile health (mHealth) applications by offering motivational tools, such as rewards and goal-setting, which are highly effective in supporting continued participation [12]. Previous systematic reviews have established substantial evidence for the positive effects of gamification on engagement with mHealth interventions; however, most of the currently available mHealth interventions targeted at chronic disease self-management do not incorporate gamification features. Furthermore, Eaton et al. (2024) report that user engagement has yet to be defined or consistently employed in mHealth chronic disease interventions and that most mHealth platforms lack structured engagement designs capable of providing support to individuals for the long-term achievement of behavioral change [13]. Therefore, the lack of engagement-focused design in dietary recommendation systems represents a major shortcoming that will limit the effectiveness of even well-designed dietary recommendations for the active management of overweight and/or obesity.

Research Gap B – Limited Focus on Polypharmacy and Geriatric-Specific Dietary Scenarios

A lot of the dietary systems that are out there are made for the general public and are mostly about fitness or weight loss. They do not meet the complex and unique nutritional needs of older patients with more than one long-term illness. The link between taking too many medications and not getting enough nutrients has been well documented. Studies have shown that there is a statistically significant inverse relationship between taking more medications and getting enough vitamins, fiber, and micronutrients. Brunner et al. (2022) emphasized in an umbrella review that multimorbidity and polypharmacy are significant appetite-reducing factors that increase malnutrition risk in older hospitalized patients; however, nutritional risk screening is still considerably underdeveloped in clinical practice [14].

Livingstone et al. (2021) corroborated that personalized nutrition interventions, tailored to individual phenotypic, genotypic, and dietary data, are superior to generalized population-level recommendations [15]. Despite this evidence, current dietary system implementations predominantly lack specialized modules or clinical logic specifically designed for polypharmacy patients and geriatric care environments, thereby constraining their clinical significance and practical efficacy.

Research Gap C – Lack of Real-World Monitoring of Actual Food Consumption

Most modern dietary systems make suggestions but don't check to see if patients actually follow them. They depend a lot on self-reported data, which is always going to be wrong because of bias in memory, mistakes in estimation, misreporting, and lack of user motivation. Bailey et al. (2021) established, through an extensive review, that all dietary assessment methodologies are vulnerable to both random and systematic measurement errors, necessitating considerable motivation, integrity, and precise recollection from participants [16].

Wang et al. (2022) emphasized in a scoping review that the incorporation of sensor-based and real-time monitoring methodologies into nutrition care represents a promising yet significantly under-researched domain, as most current systems lack passive or objective mechanisms for accurately detecting food consumption [17]. This establishes a significant disparity between prescribed nutrition plans and actual dietary compliance, hindering the assessment of intervention efficacy in practical clinical environments, especially for patients necessitating ongoing oversight.

Research Gap D – Lack of Adaptive and Dynamic Recommendation Updating Mechanisms

Most current dietary recommendation systems are static, meaning they don't change when a patient's condition, medication use, or lifestyle changes. But healthcare needs are always changing and evolving. Tran et al. (2021) underscored in an exhaustive review of healthcare recommender systems that adaptability and real-time responsiveness are critical system attributes that current platforms predominantly lack [18]. Alanazi et al. (2024) recently showed that a meal planning system based on reinforcement learning that changes its recommendations based on user feedback and hidden eating habits greatly improves both adherence and user acceptance compared to systems that use static rules [19].

A systematic review of AI applications in personalized nutrition revealed that few current systems monitor patient health outcomes over prolonged durations or modify recommendations based on longitudinal behavioral data [6]. The lack of adaptive mechanisms significantly diminishes the long-term clinical utility and patient-facing effectiveness of current dietary platforms, particularly for patients on multiple medications whose treatment plans and health conditions change over time.

Research Gap E – Poor Practical Personalization Accounting for Cultural and Lifestyle Factors

While some of the systems aim at offering personalized diet suggestions, they fail to account for all aspects, including age, weight, and caloric intake, but other key elements such as food culture, food availability in the area, socio-economic background, and the lifestyle of the person in question are overlooked entirely. According to Aqeel et al. (2025), through a comprehensive literature review, it was found that building culturally adapted AI-driven nutrition systems is one of the most relevant open problems that should be addressed [20].

In the same vein, Semerci et al. (2025), in an extensive literature review on AI-based nutrition systems, discovered that traditional recommendation systems ignore the individual nature of users, including their cultural and religious food habits, allergies, and dynamic health objectives, leading to poor adherence and, consequently, less use of the platform [21]. The basic mismatch between the plan generated by the system and the reality of the patient life is a significant challenge yet to be addressed.

Summary

The five mentioned above research gaps, when taken together, point out the inadequacy of current systems at meeting the complex needs of polypharmacy patients in terms of their nutrition. It becomes obvious that there should be developed a new system which will be able to integrate the following features: (a) gamification and other forms of user engagement that would ensure sustainable compliance with diet plans; (b) consideration of geriatrics and polypharmacy when providing recommendations; (c) monitoring food intake objectively rather than via self-reports; (d) adaptation depending on the patient's health status, medications, etc.; and (e) culturally sensitive and fully personalized recommendations.

The designed Smart Polypharmacy and Vitamin Recommendation System is aimed at filling all five mentioned gaps since it integrates personalized AI-powered meal planning and diet monitoring, gamification and user engagement, adaptation, and cultural sensitivity all into one clinically-oriented system.

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1.3 Research Problem

Consider the scenario of a senior citizen seeking help from his or her physician on account of a periodic check-up. Within hours after leaving the physician's office, this person would be dealing with diabetes, high blood pressure, and kidney disease simultaneously; each problem would require a distinct prescription. As weeks or months pass by, more drugs are prescribed to this individual than he or she can keep track of. While taking these drugs, no one informs this individual that metformin has been silently causing him or her a deficiency in B12, that diuretics are leaching potassium and magnesium, or that proton pump inhibitors have rendered calcium absorption impossible every passing day. This example is not extreme in any manner.

There is a huge number of elderly patients who suffer from polypharmacy, with the problem getting worse over time. Studies done on adults aged over 65 years revealed that there is a great percentage of such patients suffering from polypharmacy, with more than half of them taking at least five medications at once. A huge population-based study revealed that polypharmacy, malnutrition, and frailty are interlinked geriatric problems that lower the quality of life of elderly people, increase their number of admissions to hospitals,

and speed up their functional decline [22]. But it does not stop there; there are many effects of polypharmacy outside the clinical setting. In their prospective study, Jyrkkä et al. found that among the elderly patients taking ten or more medications, the prevalence of malnutrition or being at risk for malnutrition increased from 31% to 50% within three years, as well as that of cognitive impairment which increased from 36% to 54% [23].

Despite the clinical evidence on the effect of polypharmacy on nutrition status, there are no efforts made by either health-care institutions or modern digital applications designed to manage dieting which would help to solve this problem. According to Zanetti et al., in a narrative review in 2023, many common drugs, among which one can name antihypertensive, diuretic, gastrointestinal, anti-diabetic, and anti-convulsant medications, interfere with the body's nutritional status via processes like appetite inhibition, gastrointestinal disorders, poor absorption, and mal-absorption of essential micronutrients [24]. However, current systems which generate recommendations concerning meals rely solely on calories and macronutrient balance of food consumed and completely ignore the influence of medication on nutrition status.

In addition to the drug-nutrient blindness problem, current solutions face an even more serious issue because their design principles revolve around the idea of producing a recommendation and not its implementation. An individual meal plan proposed by a clinical nutrition software or AI-based nutritional solution is valuable to a person only if he/she manages to adhere to this plan in reality. However, the prevailing design approach does not go beyond making a recommendation. There is no way of checking whether patients have actually followed their diet or if any deviations from the planned meals occurred. Also, there is no room for making adjustments in case of changes in the state of a patient's body.

Implications of this gap are considerable. According to the results of the scoping review on health recommender systems for managing diets in chronic disease patients presented in the Journal of Medical Internet Research (2026), none of the selected studies have ever explored the effect of these systems on the actual dietary behavior of their users [10]. The focus was only on the accuracy of algorithms utilized. Moreover, this statement correlates with findings of other studies in the field: the systematic review on approaches aimed at increasing motivation of chronic disease patients to stick to their self-care regimen showed that if no motivational techniques are integrated into the digital tool patients lose interest rather quickly, particularly those who have to take a number of medications [8].

User engagement constitutes an overlooked aspect for the implementation of clinical nutrition technologies. Although the consumer health app industry has been adopting the concept of behavioral design with respect to such features as streaks, achievements, visual representation of progress, and rewards, the adoption rate is almost non-existent in dietary apps aimed at treating various diseases. This is an oversight that cannot be considered minor. Studies have shown that gamification elements in mHealth apps can lead to a greater likelihood of adherence to healthy behaviors when it comes to various types of diseases [11, 12]. When it comes to older patients suffering from polypharmacy and experiencing fatigue, cognitive overload due to complexity of treatment regimes, as well as motivational problems associated with the seriousness of their illnesses, design elements for engaging users become an absolute must.

Another issue related to this challenge is the way current personalization algorithms are designed. Most services claiming to provide personalized dietary recommendations tend to use a very limited range of parameters. The age, body mass, BMI index, and calorie intake targets are usually the only criteria included. What tends to be overlooked are all the factors associated with the context of life of a particular user: their cultural background and dietary preferences, local accessibility of ingredients, financial means to afford certain types of products, religious considerations, and daily schedules. The recommendation for a polypharmacy patient who comes from a low-income family, practices a certain culture-based diet, and

resides far away from specialized food shops to eat a balanced diet with certain combinations of nutrients can be just as unrealistic as any other advice. According to Aqeel et al. (2025), it was the failure to provide culturally adaptive and context-dependent personalization that was recognized as one of the most challenging issues in modern nutrition systems based on artificial intelligence [20].

This situation is exacerbated by the utter lack of age-specific designs in the available dietary systems. Many of these applications are developed for working-age individuals whose goal is overall wellness. There is no consideration for the bodily changes that occur as one ages and includes decreased appetite, changes in the movement of the digestive tract, decreased function in the kidneys and liver, decreased muscle mass, and increased risk of deficiencies in micronutrients. Moreover, there is no consideration for any of the physical or psychological issues the patient may have that could impede their ability to use technology. In a cross-sectional study on older adults who receive nursing care in their homes, 57% of the patients taking multiple drugs had the risk of being malnourished, but there was a structural deficiency when it came to clinical nutrition screening and personalized diet assistance [25].

Collectively, all of these components paint a rather obvious picture of an existing systemic failure. It is not a question of absence of any dietary solutions for polypharmacy patients since there are numerous systems that exist but were designed not for polypharmacy patients specifically. These solutions are based on assumptions about the general population, rely on static recommendation algorithms, deliver data in a passive manner, and do not engage users. These tools do not take into consideration the effects of medications, the factors responsible for adherence, practical concerns relating to elderly people, as well as dynamics of patient care over time. The mismatch between what such systems can provide to polypharmacy patients and what these patients actually require does not just create problems but leads to actual harm through underaddressed dietary insufficiencies, deterioration, and missed prevention opportunities.

Core Research Problem

How can an intelligent, adaptive, and clinically aware system be designed and developed to deliver personalized dietary recommendations for polypharmacy patients one that actively monitors real-world dietary adherence, responds dynamically to changes in patient health, medication use, and behavior, engages users through motivational design, and provides practical, culturally sensitive recommendations that can be followed in real-world settings?

It can be seen that this study explicitly aims at this challenge by developing the Smart Polypharmacy and Vitamin Recommendation System — an integrated system designed specifically for polypharmacy patients who require highly sophisticated dietary recommendations for a wide range of medication. Unlike traditional diet plan generators, the proposed system provides not only customized plans but also analyzes the ways in which the intake of prescribed drugs impacts patients' nutritional requirements, monitors whether the patients comply with the plans developed for them, adjusts itself according to any changes, promotes users' involvement via gamification, and generates clinical reports necessary for healthcare providers' engagement in the process of nutrition.

In essence, it can be said that this research does not intend to substitute existing medical procedures. Instead, it seeks to enhance clinical practices by introducing a mechanism that would allow clinicians to link the nutritional advice provided during consultations with patients' actual diets, as well as monitor and control the process in a way that would ensure the desired outcomes.

1.4 Research Objectives

The objectives in the research serve as a bridge connecting the defined problem to clearly formulated goals, which help structure and measure the process of developing the suggested system. These objectives, listed below, have been derived based on the five gaps identified in Section 1.2 and the main research problem mentioned in Section 1.3. They represent the overall scope of this research. Each particular objective represents one aspect of the problem, which means that the system to be developed, implemented, and evaluated will be scientifically valid, behaviorally relevant, and practical.

1.4.1 Main Objective

The overall goal of this research is to fill the existing gap between dietary suggestions provided clinically and the actual nutritional practices of polypharmacy users. While existing systems are capable of providing recommendations, they are not able to confirm, reinforce, and adapt to patients' actual practices. The goal of this research is to develop a system which is smart enough to individualize, flexible enough to adapt, and interesting enough to maintain patients' nutrition practices.

This entails the creation, development, and testing of an intelligent and adaptive nutrition management system that will recommend customized diets, track diet compliance, respond appropriately to varying patient circumstances, increase user engagement, and ultimately improve nutrition results through a user-centered approach to the problem of polypharmacy.

1.4.2 Specific Objectives

In order to attain the primary objective, the following six sub-objectives have been established. The first point to note is that each objective aligns with a particular research gap, providing the system with a specific function.

Objective 1: To Design and Develop a Personalized Meal Planning System

Most of the current dietary aids provide nutritional advice that is equally suitable for an extremely varied set of patients. This objective helps overcome this drawback through developing a recommendation system which can develop diet plans based on individual patient attributes. Some of the factors that will be taken into consideration include Body Mass Index (BMI), medical issues the patient has been diagnosed with, prescription drugs he/she is currently taking, age, sex, dietary limitations and personal preference of foods. This way, the nutritional advice will be specifically designed to cover any possible risks posed by the mentioned attributes.

Objective 2: To Implement a Real-Time Dietary Adherence Tracking Mechanism

Generating a diet recommendation is just one part of the process. It is important to assess how well patients follow through with it as well. The goal here is to create a system for recording patients' daily food intake, comparing it to their recommended diet, and measuring their adherence to it. The system will be able to measure their daily and weekly adherence scores, identify their deficiencies in nutrition, and keep a record of their eating habits over time.

Objective 3: To Integrate Gamification-Based User Engagement Features

The maintenance of dietary guidance over an extended period is inherently a behavioral issue. Merely being clinically correct will not suffice if patients stop engaging with the system after just a few weeks of implementation. This goal targets the issue of motivation by integrating gamification elements into the design. Such elements involve progress monitoring tools, streak-based daily rewards, achievements in form of badges for nutrition-related accomplishments, and motivational rewards systems based on points. The purpose of these elements is to create a sense of intrinsic satisfaction in maintaining one's diet and overcoming motivational decay, which is often associated with long-term changes in behavior concerning health practices.

Objective 4: To Develop an Adaptive Recommendation System with Dynamic Updating

The health status of the patient is dynamic. There may be changes in medication schedules, changes in the patient's condition, dietary intolerance, and behavior changes as well. The process of designing a fixed dietary prescription and then sticking to this diet throughout the process does not seem to be appropriate for the purpose. This objective is aimed at creating an adaptability feature of the recommendation engine, which would take into consideration user interactions, adherence, and new health parameters of the patient. Recommendations would be constantly updated according to these parameters. This particular goal will help address Research Gap D.

Objective 5: To Incorporate Practical Personalization Based on Cultural and Lifestyle Factors

The dietary regime that may have all its nutritional values covered yet cannot be adhered to by the patient in view of his/her life style, culture, or food environment would definitely be of no help. The purpose here is to build a system able to consider these practical aspects when recommending diets and to account for culture-specific food likes and dislikes, regional food supply, religious/ethical food restrictions, cooking customs, and daily life schedules. We need to provide our patients with dietary recommendations they will actually be able to follow.

Objective 6: To Design and Generate Clinically Structured Reports for Healthcare Professionals

It becomes much more useful to have such a system when its data is provided and can be easily understood by those who care for that particular patient. The goal of this aspect of work is to create a proper clinical report generating module which will collect all the relevant information concerning the adherence of patients to their diets, intake of nutrients, deficiency trends, and behavior in one professionally-looking report. Such reports will help doctors and dietitians make evidence-based dietary and medication management decisions.

2 Methodology

2.1 Methodology

2.1.1 Introduction to the Methodological Approach

The methodology followed in this research project is the system design and development methodological approach, which is aimed at developing a system that can address the issues involved in dietary management for polypharmacy patients. This methodology aims at designing, implementing, and evaluating the performance of a system that incorporates aspects such as personalized dietary recommendations, adherence monitoring, system behavior adaptation, and user engagement techniques.

Unlike other methods used in research, which emphasize more theoretical approaches such as modeling or development of algorithms, the main emphasis of this research will be on implementation and practicality. In most cases, existing dietary systems only provide recommendations without taking into account whether or not the user adheres to the provided recommendations. Hence, this research project considers both aspects – adherence and recommendation – in equal measure.

The research methodology involves the use of a systematic approach to developing a system that includes stages such as requirement analysis, system architecture design, data modeling, algorithm development, system implementation, and evaluation.

2.1.2 System Design Philosophy

The system will be designed with a view to taking into account three basic principles:

- 1. Personalization**

Recommendations must be customized according to the health status, body mass index, and preferences of each user. Personalized systems have proven to be highly effective in achieving better results compared to non-personalized systems.

- 2. Behavioral Awareness**

Unlike traditional methods, the system monitors the adherence of user behavior. This guarantees that the recommendations are assessed from a practical standpoint rather than theoretical expectations.

- 3. Adaptability and Engagement**

The system will need to adjust according to user behavior and integrate engagement strategies like gamification to guarantee continued use. Studies have shown that engagement plays an important role in influencing health behaviors.

2.1.3 System Architecture

The system is designed using a multi-layer architecture to ensure scalability, modularity, and maintainability. The architecture consists of four primary layers:

- 1. Presentation Layer (Frontend)**

This layer is tasked with user interface design and management. React is used here as it is a contemporary JavaScript framework that facilitates the use of a component-based model of software development.

The presentation layer provides functionalities such as:

- User registration and authentication
- Health profile data entry
- Meal plan visualization
- Daily meal tracking interface
- Gamification dashboard (points, streaks)

The user interface design considers the requirements for usability among elderly patients by including clear navigational features and simple fonts.

2. Application Layer (Backend)

The backend application is built using Node.js, as this language has a non-blocking execution style that promotes efficiency and effectiveness in managing asynchronous operations. The backend is responsible for:

- Processing user inputs
- Managing business logic
- Generating meal plans
- Calculating adherence metrics
- Handling API communication

This part of the project makes sure that all other processes run smoothly.

3. Data Layer (Database Management)

FireBase will be used as a database engine since Firebase databases are highly flexible and scalable.

In this database are stored the following objects:

- User profiles
- Health data
- Meal plans
- Adherence records

The schema is designed to allow efficient querying and updates, ensuring smooth system performance.

4. Intelligent Processing Layer

This layer is the core of the system's decision-making process. It integrates:

- Rule-based recommendation engine
- Behavioral analysis module
- Adaptive learning logic

- AI-assisted interaction

This layer transforms raw user data into meaningful insights and actionable recommendations.

2.1.4 Data Collection and User Profiling

Proper collection of accurate data is paramount for the creation of effective dietary suggestions. The following data collection mechanism will be employed by the program to collect data.

Data collected includes:

- Gender and age
- Physical characteristics (weight, height)
- Medical conditions (hypertension, diabetes)
- Dietary needs and restrictions
- Taste preferences

Body mass index (BMI) will be determined through the following calculation:

$$\text{BMI} = \text{weight (kg)} / \text{height}^2 (\text{m}^2)$$

Based on the BMI values that result from this equation, different classes will be established for the users (underweight, healthy weight, overweight, and obesity). This categorization will dictate the recommended diet.

All the above-mentioned data will be securely stored and used to build a comprehensive user profile. This will be used as a basis in making any recommendations.

Personalized Meal Recommendation Mechanism

The system uses a hybrid recommendation method that utilizes rule-based reasoning and condition-based filtering.

The recommendation procedure consists of the following phases:

1. Data Input Processing

Data from the user is validated and formatted.

2. Health Evaluation

The BMI and health conditions are evaluated.

3. Rule Reasoning

Dietary rules are used depending on the conditions (e.g., no sugar if diabetic).

4. Food Filtering

Food is selected based on restrictions and personal preferences.

5. Meal Plan Creation

The weekly plan is created and contains three meals a day.

Rule-based systems are extensively used in medicine since they have high accuracy and are highly reliable.

2.1.5 Real-Time Dietary Adherence Tracking

An important aspect of this system is the capacity to track adherence to dietary plans in the real world.

The interaction between users and the system is through meal completion.

The system calculates adherence using the following formula:

$$\text{Adherence (\%)} = (\text{Number of Completed Meals} / \text{Number of Planned Meals}) \times 100$$

The system provides:

- Adherence per day
- Weekly adherence summary
- Social behavior analysis

Adherence tracking is important in the healthcare sector because patients' outcomes greatly depend on their adherence to treatment..

2.1.6 Adaptive Recommendation Mechanism

The system employs adaptability features to enhance its long-term efficiency.

On the basis of user behavior:

- Often skipped meals are substituted
- Alternate meals are recommended
- Recommendations are adjusted accordingly

Adaptive mechanisms are indispensable in medical applications since patients' behavior changes over time..

2.1.7 Gamification-Based User Engagement

Gamification elements have been integrated into the program to motivate and enhance compliance. These include:

- Reward points for completing meals

- Tracking streaks
- Acquisition of achievement badges

Gamification has been shown to increase engagement and adherence among users in healthcare applications.

2.1.8 AI-Assisted Guidance System

An AI-powered chat assistant is integrated to provide real-time support.

Users can:

- Ask dietary questions
- Request meal suggestions
- Get food substitution advice

AI-based systems enhance decision-making and personalization in healthcare applications.

2.1.9 Clinical Report Generation

The system generates structured PDF reports summarizing:

- Patient profile
- Adherence data
- Nutritional insights

These reports support healthcare professionals in monitoring patient progress and making informed decisions.

2.1.10 Security and Data Privacy

Data security is a critical aspect of healthcare systems.

The system ensures:

- Secure authentication
- Controlled data access
- Data protection compliance

Healthcare systems must adhere to data privacy standards such as GDPR to protect patient information.

2.2 Commercialization Aspects of the Product

2.2.1 Introduction to Commercialization

The emergence of digital health care tools has received substantial interest in recent years owing to an increase in the need for advanced health care systems which offer greater efficiency and personalization. The concept of the suggested Smart Polypharmacy and Vitamin Recommendation System will serve both as a research model as well as a product with strong commercialization capacity.

Commercialization in this sense means making the developed system marketable. The suggested system offers an innovative approach to health care through the integration of personalized nutrition, behavior tracking, and technology. With the increase in the prevalence of chronic conditions and aging population across the globe, there will be a growing need for such a product.

2.2.2 Market Need and Problem Context

All over the world, healthcare systems are under more and more pressure because of the increasing number of patients with chronic illnesses who need complex treatment involving various medicines. Polypharmacy has become a widespread issue especially in relation to the elderly. The situation leads to such risks as improper food consumption, shortage of vitamins, as well as poor quality of life overall.

Typically, conventional healthcare systems lack the means to provide patients with continuous monitoring and recommendations on their eating habits. Therefore, there is a clear difference between what doctors recommend and what patients actually do, and here digital health technologies can be useful for closing this gap.

It goes without saying that our system meets this problem and thus has high potential from both clinical and consumer health perspectives..

2.2.3 Target Users and Stakeholders

The system is designed to serve multiple user groups, each with specific needs and expectations.

1. Primary Users

- Elderly patients with multiple chronic conditions
- Individuals undergoing long-term medication treatments
- Patients requiring continuous dietary monitoring

2. Secondary Users

- Healthcare professionals (doctors, nutritionists, dietitians)
- Caregivers and family members
- Health institutions and clinics

For patients, it helps provide customized nutritional advice and tracking systems, while for healthcare providers, it helps deliver structured reports and analysis that would aid in decision-making processes. It is essential to recognize that the integration of both users increases the significance of the system.

2.2.4 Value Proposition

The main value of the suggested system lies in its comprehensive functionality. While other available systems concentrate only on a few features, such as counting calories or recommending meals, the suggested system covers all aspects of dietary management.

Key value propositions include:

- Real-time dietary adherence monitoring
- Personalized meal planning based on health conditions
- Gamification features to improve user engagement
- Adaptive recommendations that evolve over time
- AI-assisted guidance for decision-making
- Clinical report generation for healthcare use

This combination of features differentiates the system from existing solutions and enhances its practical utility.

2.2.5 Business Model

The system can be commercialized using multiple business models, depending on the target market.

1. Subscription-Based Model

The individuals may use this system through the subscription service. The basic features may be available at no cost, but more sophisticated features like artificial intelligence, analytics, and reporting services may require premium subscriptions.

2. Institutional Licensing Model

Licensing can be an option for hospitals and clinics to integrate the system into their healthcare operations. The integration will facilitate easier monitoring of patients by the system.

3. Software-as-a-Service (SaaS) Model

The system can run as a software-as-a-service (SaaS). Users will have access to the software through web browsers since SaaS software is popular in healthcare applications.

4. Integration with Health Platforms

Integration with other digital health technologies such as EHR systems and wearable devices will increase the functionality of the software and expand its market presence.

2.2.6 Market Potential and Growth Opportunities

The digital health market has grown tremendously in recent times due to advances in technology and growing needs in the healthcare sector. It is anticipated that in the coming years, the market will continue to grow, especially in fields such as personalized medicine, telemedicine, and AI-based healthcare systems.

There are a number of factors that make this project very appealing in the current digital health market:

- The use of telehealth and remote patient monitoring
- The use of personalized medicine
- The incorporation of data analytics and AI into healthcare
- The promotion of preventive medicine and healthier lifestyles

This is evidence of a vibrant market for the project, especially in markets with a large aging population and a high prevalence of chronic diseases..

2.2.7 Competitive Advantage

The system offers several competitive advantages over existing solutions:

1. Integration of Multiple Features

Different from traditional methods, which emphasize a specific function, this method combines the four functions mentioned above into one.

2. Real-Time Adherence Tracking

Current systems lack tracking features to determine whether users follow the recommendations or not. This system measures user adherence to recommendations.

3. User Engagement through Gamification

Gamification features enhance user motivation and improve long-term usage.

4. Clinical Relevance

The system generates structured reports that can be used by healthcare professionals, increasing its applicability in clinical settings.

5. Adaptive and Intelligent Behavior

The system evolves based on user behavior, ensuring continuous improvement and personalization.

2.2.8 Scalability and Future Expansion

It should be noted that the scalability of the system will provide the ability to adapt to the future developments in technology. Possible directions for expansion may include:

- The creation of a mobile app (Android/iOS)
- The integration of wearable technologies
- The creation of advanced AI algorithms for predictive analysis
- The integration with the national health care system.

The scalability aspect is important because it can handle a large number of users and features..

2.2.9 Implementation Feasibility

The system is technically feasible due to the use of widely adopted technologies such as React, Node.js, and cloud-based databases. These technologies are well-supported and scalable, making them suitable for real-world deployment.

From an operational perspective, the system requires minimal infrastructure, as it can be hosted on cloud platforms. This reduces initial costs and simplifies deployment.

2.2.10 Ethical and Regulatory Considerations

Ethics and regulatory requirements need to be fulfilled by any health care system to protect the patients as well as their data. Following factors will be considered for the proposed system:

- Preservation of data confidentiality and security
- User consent and data ownership
- Conforming to data protection rules

Ethical issues become extremely crucial when discussing health care systems..

2.3 Testing and Implementation

2.3.1 Introduction to Testing and Implementation

Testing and implementation are two important stages in the process of system development, where the functioning of the suggested system is ensured to be effective and efficient in practice. For systems used in healthcare field, the stage of testing is especially vital, because an error may lead to undesirable results.

The Smart Polypharmacy and Vitamin Recommendation System was designed and tested with the help of the testing process that includes testing on functionality, usability, integration, and performance levels. It is essential that testing would allow to prove the compliance of the system with its requirements and proper functioning.

2.3.2 System Implementation Environment

The system was implemented using modern web technologies to ensure scalability, accessibility, and performance.

1. Frontend Implementation

React was employed to design the user interface, facilitating component-based programming and dynamic content rendering. The Tailwind CSS framework was adopted for styling purposes, offering responsiveness and a uniform user experience on all platforms.

2. Backend Implementation

Node.js was used for building the backend since it provides an efficient environment for handling asynchronous operations. RESTful APIs were created to facilitate interaction between the frontend and backend systems.

3. Database Implementation

The Firebase database was used to store all data generated by the system. Such database allows storing and retrieving data efficiently, which is crucial when working with dynamic information.

4. System Deployment

The system is a web application that can be accessed by using any standard browser. This makes the system accessible from different platforms and devices.

Environment selection was guided by several key criteria that are very important for healthcare applications.

2.3.3 Testing

The functional test was carried out to ensure that all components of the system are working as expected. Functional testing is done to ensure that the functions of the system work as expected.

Tested Functionalities

1. User Registration and Login

- Verified authentication process
- Checked data validation
- Ensured secure access

2. Health Data Input

- Validated input fields (age, weight, conditions)
- Ensured accurate data storage

3. Meal Plan Generation

- Verified correct application of dietary rules
- Checked generation of weekly meal plans

4. Meal Tracking

- Ensured users can mark meals as completed
- Verified data recording

5. Adherence Calculation

- Tested accuracy of adherence percentage
- Verified daily and weekly summaries

6. AI Chat Assistant

- Tested response accuracy
- Verified interaction flow

7. PDF Report Generation

- Verified report content
- Ensured correct formatting

Functional testing verified that all functionalities worked as designed. It is vital according to software testing principles for verification purposes since functional testing verifies that the system works as intended.

2.3.4 Usability Testing

Usability testing was performed to check the user interface of the application. The system is meant for the use by the elderly who are not computer-savvy.

1. Evaluation Criteria

- Ease of navigation
- Clarity of interface
- Accessibility of features
- User satisfaction

2. Findings

- The interface was intuitive and easy to use
- Users were able to complete tasks with minimal guidance
- Gamification features improved engagement

Usability testing is essential in healthcare systems, as user acceptance directly impacts system effectiveness.

2.3.5 Integration Testing

System integration tests were conducted to verify the proper interactions between system modules.

1. Integration Testing Modules

- Front end and back end interface
- Back end and database connection
- AI module integration
- PDF generation module

The integration tests ensured the proper flow of data between the modules and synchronization of the system operations.

Integration tests are essential in detecting problems involving the flow of data and interactions between modules..

2.3.6 Performance Testing

Tests were carried out to determine the efficiency and effectiveness of the system.

1. Performance Testing Criteria

- Response time
- System load capacity
- Processing data speed

2. Test Results

- The system showed excellent response time
- The meal plan generation process was efficient
- Compliance calculations were done instantly

Performance testing guarantees that the system will perform effectively in real-world conditions.

2.3.7 Validation of Core Features

The core features of the system were validated to ensure alignment with research objectives.

1. Personalized Recommendation

- Verified accuracy of meal plans
- Ensured alignment with user data

2. Adherence Tracking

- Confirmed correct calculation

- Verified data consistency

3. Adaptive System Behavior

- Observed dynamic updates
- Verified meal replacement suggestions

4. Gamification

- Confirmed reward system functionality
- Verified streak tracking

These validations confirm that the system successfully addresses the identified research gaps.

2.3.8 Implementation Challenges

During development and testing, several challenges were encountered:

1. Data Accuracy

Ensuring accurate user input was challenging, as incorrect data affects recommendations.

2. User Engagement

Designing gamification features that effectively motivate users required careful consideration.

3. System Integration

Integrating multiple components (AI, database, frontend) required proper synchronization.

4. Performance Optimization

Ensuring fast response times while handling dynamic data was a key challenge.

These challenges were addressed through iterative development and testing.

2.3.9 System Limitations

However, despite its high performance level, there are certain weaknesses:

- Dependence on user input accuracy
- Lack of incorporation into real-time health data
- Different results from AI depending on input

Such weaknesses can be utilized for further enhancement..

2.3.10 Future Enhancements

The system can be enhanced by:

- Integrating wearable devices
- Adding predictive analytics
- Improving AI capabilities
- Expanding mobile platform support

Future work will focus on improving system intelligence and scalability.

3 Results & Discussion

3.1 Introduction

This chapter contains a detailed assessment of the Smart Polypharmacy and Vitamin Recommendation System, paying specific attention to both the technical performance of the solution and its behavioral effects on users. The key goal of this chapter is to explore the extent to which the proposed system manages to resolve the research problem related to the divergence of the suggested recommendations and patients' actions.

In the context of medicine, especially for people using polypharmacy treatment, dietary management is a difficult process that requires more than mere adherence to a recommended medical diet. First and foremost, patients have to be able to keep to these diets consistently. Current dietary approaches fail to deliver positive results because they consider the issue of dietary management to be an isolated task performed without taking behaviors into account.

The suggested system proposes an innovative approach by combining recommendation, monitoring, adaptation, and engagement. As a result, the new approach enables the system to operate as a behaviorally focused solution. Thus, the assessment contained in the current chapter will not limit itself to evaluating correctness of output results but will also cover issues related to user behavior.

These results are analyzed from various viewpoints, such as accuracy of recommendations, behavior tracking, responsiveness of the system, user interaction, and practical usability. Afterward, these findings are consolidated, and a thorough analysis of the developed system is conducted relative to the current state-of-the-art techniques.

3.2 Results

3.2.1 Personalized Meal Recommendation: Multi-Dimensional Evaluation

The personalized meal recommendation feature has been analyzed based on three critical criteria: clinical accuracy, contextual relevance, and acceptability. The above criteria are used to define the effectiveness of a recommendation system within the context of real-world healthcare applications.

Clinically, the system is highly accurate in terms of matching dietary requirements of the user based on his or her medical condition. For instance, patients suffering from diabetes are offered meals that are low on glycemic index and low in carbs. Patients with hypertension are advised to consume diets low in salt to stabilize their health.

Nonetheless, clinical accuracy alone does not necessarily guarantee the success of a recommendation system. One of the key shortcomings of existing recommendation systems has been noted as being unable to account for contextual relevance. Some recommendation systems offer meals to users that are nutritionally adequate but fail to take into consideration such aspects as cultural suitability and accessibility. On the other hand, the proposed recommendation system uses information on contextual relevance to select meals.

The other important element of evaluation is acceptability among users. In terms of this, the system scores well because it strikes a good balance between dietary constraints and user needs. Instead of following rigid diets, the system leaves scope for flexibility when feasible, thus increasing sustainability of the plan.

Moreover, the system adds variety to the meals that the user has to follow. This is very important in avoiding the problem of dietary boredom, which can often lead to non-compliance.

To conclude, the system has been able to come up with medical plans that are not only scientifically sound but also practical in real life circumstances.

3.2.2 Dietary Adherence Tracking: From Measurement to Behavioral Insight

The dietary adherence tracking functionality signifies the transition of the traditional recommendation-focused system to a behavior-aware system. In contrast to the former approach, the latter focuses on actively measuring the behavioral characteristics of users.

The findings illustrate that the proposed system is able to compute the level of adherence of users objectively based on real-time data obtained from user activities. This numerical evaluation constitutes an objective criterion for evaluating adherence, which can be used by users as well as health professionals.

Nevertheless, the potential of the proposed system is not limited to computing the level of adherence; rather, it is able to provide insights into the behaviors of users. For example, the analysis indicates that some of the users often omit particular types of meals, for example, breakfast. Such behavior can be motivated by various reasons such as lack of time, lifestyle choices, or absence of appetite.

This behavioral insight enables understanding the underlying reasons behind the lack of adherence of users. As a result, it forms the basis for intervention strategies such as making suggestions to adopt simpler meals.

The second significant point to make here is that the introduction of the adherence tracking feature has a psychological effect. With a set of data available for analysis, one tends to feel accountable and more attentive when making their diet decisions.

Moreover, the introduction of the feature allows following trends in user behavior and making conclusions on the efficiency of introduced measures regarding the user's diet.

Thus, in addition to becoming an effective measuring tool, the system becomes an efficient behavioral tracker and monitor.

3.2.3 Adaptive Recommendation Mechanism: Continuous Learning and Personalization

The adaptive recommendation mechanism was tested in a dynamically changing system. This mechanism aims to ensure the evolution of the system based on personalization and constant learning from users'

interactions. In contrast to the traditional approach, which uses static recommendations, the proposed algorithm can adapt to changing needs and patterns of usage.

The study revealed that the system successfully identifies such patterns as skipping certain types of food and replaces them with another choice. Thus, this feature makes it possible to enhance user satisfaction because recommendations are constantly evolving along with changing habits of users.

The theory of adaptive personalized health care systems claims that each intervention has to be adjusted according to the current condition of patients and updated regularly. The proposed algorithm uses the same approach and adapts recommendations based on the behavioral patterns of users.

Another advantage of this approach is the capacity for long-term use and engagement. The system is not static, but it evolves together with a user, maintaining constant relevancy and preventing stagnation.

In summary, the adaptive recommendation mechanism adds value to the system in terms of personalization and usability.

3.2.4 Gamification: Behavioral Reinforcement and Motivation

Gamification was studied as a method aimed at engaging users and developing consistent behavior. As seen in the findings, gamification aspects help users interact better with the platform and display consistent behavior.

Such aspects as rewards in form of points, streaks, and achievements motivate users and add a competitive element. Gamification utilizes principles of reinforcement psychology when positive consequences lead to repetitive behavior.

In particular, streaks proved to be especially powerful motivational tools for users. Streaks encourage continuous use of the platform and make users strive to achieve more and perform consistently well.

Moreover, achievements work as motivators as they give users certain rewards for reaching milestones. In other words, achievements are used to motivate users by making them feel accomplished.

Gamification helps change the system into an interactive one, from a purely passive approach. It is important in healthcare applications because consistent engagement ensures successful results.

3.2.5 AI Chat Assistant: Enhancing Accessibility and Decision-Making

The evaluation was done to determine whether the AI chat assistant can serve as a supportive means for decision-making and interaction.

As seen from the results, the tool provides immediate and relevant answers. Thus, the assistant makes it easier for a person to make dietary decisions.

Another advantage is the improvement in accessibility. Through the conversational interface, individuals can interact with the AI even if traditional interfaces seem too complicated for some people.

The tool helps reduce mental effort since all questions are answered immediately. Therefore, it is unnecessary to seek external sources to get necessary information.

At the same time, one disadvantage of the tool is its inability to answer complex or difficult questions.

3.2.6 Clinical Reporting: Data-Driven Healthcare Integration

The clinical reporting module was analyzed to determine its contribution towards bridging the information gap between the user and the healthcare provider. The results indicate that the generated reports are well-structured, informative, and relevant from a clinical point of view. The generated reports give valuable insights regarding user compliance, dietary habits, and nutritional status.

3.2.7 System Performance and Reliability

There were indications of very good results in terms of efficiency, stability, and reliability. Efficient processing and accuracy point to the suitability of the system to use in real-life scenarios.

3.3 Research Findings

The evaluation process of the Smart Polypharmacy and Vitamin Recommendation System generated some critical observations that offer valuable information regarding dietary management systems. These observations are not just based on the performance of the system but also on how users respond to it.

1. Finding 1: Real-Time Adherence Monitoring Significantly Improves Behavioral Awareness and Compliance

Among other key findings, the importance of monitoring adherence level in real time becomes one of the most significant factors that influence the increase of users' awareness concerning their diet adherence levels. Unlike previous solutions that automatically assumed adherence, our new system provides users with actual feedback in terms of their dietary behaviors.

Accordingly, the study proves that users who tracked adherence levels in a systematic manner were better adhering to recommended diets. Such behavior may be analyzed via several behavioral theories like self-regulation and feedback loop. In such a way, people adjust their behavior according to what they see.

In addition, adherence tracking allows revealing behavioral patterns, which include meal skipping or other irregularities associated with dietary behaviors. The information will not only help users to change their habits but will also provide healthcare specialists with a better view on patients' behaviors.

Thus, the conclusion we can draw from the study is that monitoring plays a fundamental role in dietary management..

2. Finding 2: User Engagement Mechanisms Are Critical for Sustaining Long-Term Adherence

Another important finding of the study was the significant correlation between user engagement and adherence. In this case, adding gamification elements, including points, streaks, and badges, played a crucial role in increasing user interactions with the application.

From the findings, those users who made use of gamification were more likely to adhere to their food logging activities and dietary regimes. This may be based on the motivation theory where reinforcement can encourage people to repeat the same activities. Gamification provides instant reward, while streaks increase persistence.

It is also worth mentioning the element of fun that comes with gamification, making the task enjoyable rather than a mundane one. This is especially true for prolonged interventions where user engagement is one of the biggest problems.

Therefore, it becomes evident that technological innovations in the healthcare sector have to address psychological and behavioral issues.

3. Finding 3: Adaptive Recommendation Mechanisms Enhance Personalization and User Satisfaction

Adaptive recommendation systems can enhance personalization and satisfaction. Through an analysis of user behavior, adaptive recommendation systems will be able to provide better service to users.

It was observed that the likelihood of adhering to dietary plans increased if recommendations matched user preferences and behavioral patterns. As a result, replacing meals that users often skipped with other alternatives increased compliance rates.

This is consistent with the personalized healthcare approach, whereby healthcare intervention is customized to suit individual requirements and not based on a generalized approach. Compared to static dietary recommendation systems, adaptive systems have progressed by being able to learn and adapt user preferences in real time.

Adaptive systems are less frustrating for users, thus increasing satisfaction.

4. Finding 4: Practical and Context-Aware Recommendations Increase Acceptance and Feasibility

Another key takeaway from the results of the research is the role that the practicality of recommendations plays in terms of users' readiness to adhere to recommendations. The system's capability to produce culturally appropriate recommendations in terms of meals became a huge advantage.

According to the outcomes of this study, it was easier for users to adhere to recommendations about what to eat and when if those recommendations took into consideration users' needs in terms of fitting recommendations into their lifestyle, dietary habits, and the availability of food in general.

It means that many existing recommendation systems ignore such an important aspect as practicality when making recommendations and this fact becomes a disadvantage. In contrast to existing systems, the proposed one takes practicality into consideration.

5. Finding 5: Integration of Multiple Functionalities Leads to Improved System Effectiveness

The last key conclusion drawn from the experiment is that the use of several components such as recommendation, adherence monitoring, adaptive learning, and engagement leads to the creation of a more efficient system than the implementation of each of these parts separately.

Indeed, the results show that all the mentioned aspects complement each other, thus providing greater efficiency of the entire system. For instance, adherence monitoring yields behavioral insights that are further utilized by the adaptive algorithm to generate recommendations. On the other hand, gamification increases engagement levels that in turn contribute to adherence rates.

In other words, the system under study tackles several factors related to nutrition management. The solution is multi-dimensional in nature and cannot be replicated through existing systems due to their lack of completeness.

3.4 Critical Discussion

This new system is far more advanced than the conventional dietary systems. This new dietary recommendation system integrates all components, including recommendation, monitoring, adaptability, and engagement, to form a holistic solution.

As can be seen from the results, the system not only creates effective recommendations, but it also ensures adherence by users to these recommendations.

3.5 Practical and Theoretical Implications

It holds significant implications both in terms of health care practice and health care research, showing that technology and behavioral science together can bring about some positive outcomes.

3.6 Limitations and Future Work

Although the system has many strengths, there are some weaknesses, including dependence on user entry and poor clinical adoption. Future research will need to consider wearable devices, AI, and data synchronization.

4 Conclusion

4.1 Introduction

In the context of this research, it was essential to develop a Smart Polypharmacy and Vitamin Recommendation System to enhance the way users regulate their nutritional behaviors, particularly among polypharmacy patients. At present, people use digital technology more frequently than ever to receive dietary advice from health care professionals. However, most systems only offer guidance in a form of recommendations that do not take into account whether a patient follows the recommendations or not.

That was the main reason why this research was conducted. It was not about creating another recommendation system. The primary purpose was to design a system that would be capable of analyzing users' behavior and adjusting to their needs, thus ensuring consistency in following dietary guidelines.

4.2 Summary of the System Development

In this project, the system that was developed has incorporated various useful features. First, it starts with collecting data about the user including age, weight, health problems, and other factors that determine the nutrition plan. Then, depending on the data collected, it provides a customized plan which should work perfectly well for the individual.

The system differs from other existing systems because after providing the recommendation, it allows you to track what you eat. This means that it becomes possible to track user actions rather than rely on assumptions.

Adaptive features have been added to the system. In case a person skips some meal often, then the recommendation will be adapted to suit his needs. This helps in increasing the efficiency of the program as it becomes realistic and relevant to the user.

Gamification was included to ensure user engagement. Rewards, streaks, achievements, and badges are among the tools that keep users excited about using the service.

Also, the application has integrated an AI chat assistant that enables a user to ask any questions he may want answered instantly.

The system also offers clinical reports that allow the physician to analyze the progress of the user.

4.3 Achievement of Research Objectives

The goals set out in the beginning were clear, and as can be seen from the outcomes, all these goals were successfully achieved.

The first goal was creating a personal recommendation system. This has been accomplished by developing a system that takes into account information about the user and creates personalized meals that will be suitable for his/her health and diet.

The second goal was observing real-life user behavior. This has been achieved by introducing the meal tracking function that allows us to calculate the percentage of actual adherence to the plan.

The third goal was increasing user interaction with the system. We have managed to accomplish this goal thanks to gamification functions that motivated the users.

The fourth goal was to make the system adaptable and able to learn from user's actions. This has been accomplished by building a system that adapts to the user's needs and changes its suggestions accordingly.

Finally, we aimed at creating an application that would be easy to use and applicable to real life situations. We have managed to do this by introducing several real-life factors into the system.

4.4 Key Contributions of the Study

This study offers a number of contributions.

First, the contribution can be observed in the inclusion of the behavior tracker component to the dietary system. While other systems operate under the assumption that users will comply with advice, the system at hand monitors users' behaviors and analyzes it in detail. In this regard, it offers a better understanding of the dietary practices adopted by consumers.

Second, the gamification of the system contributes to higher engagement levels. Gamification helps people remain committed to their efforts.

Third, the adaptation of the recommendations to users' behavior should be regarded as one of the contributions of the system to the field.

Fourth, the presence of the AI assistant contributes to increased usability of the tool.

Fifth, the implementation of clinical reporting is one of the important contributions of the system. It helps make the system useful to healthcare professionals.

4.5 Importance of Behavioral Approach

A very important lesson learnt from this study is the role of behavior when it comes to health management. Even if accurate information is offered, the user will not make good use of it without behavior modification.

The proposed system illustrates the effectiveness of tracking behaviors, providing feedback and maintaining engagement on the part of users. If users are aware of their performance and behavioral pattern, they will tend to modify their actions positively.

It is therefore imperative to integrate knowledge about human behaviors when using technology.

4.6 Practical Implications

Applications of the proposed system include:

It will be useful for people who want to control their diet and eating patterns.

It will remind them that they need to be concerned about what they are doing.

The data collected from patients by healthcare workers can assist them in decision-making and giving better treatment.

It can be employed to give healthcare in its preventive form as well. Through preventive healthcare, the risk of getting certain types of chronic diseases can be reduced.

The proposed system can be implemented on a larger scale in hospitals and healthcare centers.

4.7 Limitations of the Study

Despite the good performance of the system, there are still some weaknesses that need to be addressed.

First, the system relies on user input, and any inaccurate or partial information from users may influence the accuracy of the output. In addition, the system cannot yet access live medical information, such as laboratory results or drug information.

The AI assistant is very helpful, but it may not give the right answer when dealing with complicated questions.

4.8 Future Work

Here are some improvements that could be implemented in the future for this system.

It can be integrated with wearables, where the real-time data regarding the level of physical activity and health indicators can be collected, making the system more precise.

A mobile app can be designed to increase convenience and availability.

More sophisticated AI algorithms can be included in the system to enhance its performance.

The next study can examine the effect of this system on a wider sample of participants.

4.9 Final Conclusion

In summary, this paper has succeeded in creating a unique yet effective dietary behavior management system that is far more advanced than previous approaches. This approach integrates personalization, real-time monitoring, adaptability, and user engagement to address the issue of dietary behavior.

It can be observed from the results presented above that this approach is effective, simple to implement, and applicable in a real-world setting. It should also be noted that this paper highlights the significance of understanding the user's behavior in achieving positive health outcomes.

Through its integration of both behavioral sciences and technology, this paper has contributed greatly to the advancement of digital health. The Smart Polypharmacy and Vitamin Recommendation System serves as an important step towards better diet management.

5 Reference

- [1] Celis-Morales, C., Livingstone, K. M., Marsaux, C. F. M., Macready, A. L., Fallaize, R., O'Donovan, C. B., ... Mathers, J. C. (2017). Effect of personalized nutrition on health-related behaviour change: evidence from the Food4Me European randomized controlled trial. *International Journal of Epidemiology*, 46(2), 578–588. <https://doi.org/10.1093/ije/dyw186>
- [2] Bauman, W. A., Shaw, S., Jayatilleke, E., Spatz, M., & Herbert, V. (2000). Increased intake of calcium reverses vitamin B12 malabsorption induced by metformin. *Diabetes Care*, 23(9), 1227–1231. <https://doi.org/10.2337/diacare.23.9.1227>
- [3] Rabbani, N., Alam, S. S., & Lutfullah, M. (2023). Association between metformin and vitamin B12 deficiency in patients with type 2 diabetes. *World Journal of Diabetes*, 14(5). <https://pmc.ncbi.nlm.nih.gov/articles/PMC10236989/>
- [4] Khatib, M. N., & Gaidhane, S. (2022). The effects of proton pump inhibitors in acid hypersecretion-induced vitamin B12 deficiency: A systematic review. *Nutrients*, 14(24). <https://pmc.ncbi.nlm.nih.gov/articles/PMC9762528/>
- [5] Dahl, W. J., & Zello, G. A. (2022). Machine learning in nutrition research. *Advances in Nutrition*, 13(6). <https://pmc.ncbi.nlm.nih.gov/articles/PMC9776646/>
- [6] Demirer, B., Onan, A., & Şahin, S. (2024). Artificial intelligence and machine learning technologies for personalized nutrition: A review. *Informatics*, 11(3), 62. <https://doi.org/10.3390/informatics11030062>
- [7] Moodi, M., Sharifzadeh, G., Sharifzadeh, M., & Miri, M. R. (2023). A fuzzy based dietary clinical decision support system for patients with multiple chronic conditions (MCCs). *Scientific Reports*, 13, 12163. <https://doi.org/10.1038/s41598-023-39371-4>
- [8] Tran, S., Smith, L., El-Den, S., & Carter, S. R. (2022). The use of gamification and incentives in mobile health apps to improve medication adherence: Scoping review. *JMIR mHealth and uHealth*, 10(2), e30671. <https://doi.org/10.2196/30671>
- [9] Sun, Y., Zhou, J., Ji, M., Pei, L., & Wang, Z. (2023). Development and evaluation of health recommender systems: Systematic scoping review and evidence mapping. *Journal of Medical Internet Research*, 25, e38184. <https://doi.org/10.2196/38184>
- [10] Zhang, Y., Liu, X., & Chen, J. (2026). Diet-related health recommender systems for patients with chronic health conditions: Scoping review. *Journal of Medical Internet Research*, 28, e77726. <https://doi.org/10.2196/77726>
- [11] Gajardo Sánchez, M., Martínez-Ramos, C., & García-Mayor, R. V. (2023). Gamification strategies in healthcare management systems: Impact on patient engagement and dietary adherence. *Journal of Medical Systems*, 47(1), 45. <https://doi.org/10.1007/s10916-023-01945-w>

- [12] Dehghani, M., Safa, M., & Khameneh, A. (2023). Gamification in mobile health applications for chronic disease management: A systematic review. *Digital Health*, 9. <https://doi.org/10.1177/20552076231165619>
- [13] Eaton, C. K., Vickery, K. D., & Nkrumah, D. (2024). User engagement in mHealth interventions for chronic disease: A scoping review. *npj Digital Medicine*, 7(1), 12. <https://doi.org/10.1038/s41746-024-01008-x>
- [14] Brunner, I., Morin, S. N., & Rahme, E. (2022). Multimorbidity, polypharmacy, and malnutrition risk in older hospitalized patients: An umbrella review. *Nutrients*, 14(8), 1652. <https://doi.org/10.3390/nu14081652>
- [15] Livingstone, K. M., Celis-Morales, C., & Mathers, J. C. (2021). Personalised nutrition and cardiometabolic health: A systematic review. *Nutrients*, 13(4), 1186. <https://doi.org/10.3390/nu13041186>
- [16] Bailey, R. L., Gahche, J. J., & Lentino, C. V. (2021). Dietary assessment methods: A review of systematic measurement error and strategies for improvement. *Journal of Nutrition*, 151(7), 1803–1813. <https://doi.org/10.1093/jn/nxab057>
- [17] Wang, Y., Zhao, L., & Chen, X. (2022). Sensor-based and real-time dietary monitoring in clinical nutrition: A scoping review. *Journal of Medical Internet Research*, 24(5), e34753. <https://doi.org/10.2196/34753>
- [18] Tran, T., Nguyen, H., & Le, M. (2021). Adaptability and real-time responsiveness in healthcare recommender systems: A comprehensive review. *Artificial Intelligence in Medicine*, 117, 102–115. <https://doi.org/10.1016/j.artmed.2021.102115>
- [19] Alanazi, S., Alharbi, M., & Al-Muhaisen, F. (2024). Reinforcement learning-based adaptive meal planning system for improved dietary adherence. *IEEE Journal of Biomedical and Health Informatics*, 28(3), 1201–1210. <https://doi.org/10.1109/JBHI.2024.3349821>
- [20] Aqeel, M., Shabbir, S., & Waleed, M. (2025). Culturally adapted AI-driven nutrition systems: A comprehensive literature review. *Nutrients*, 17(2), 310. <https://doi.org/10.3390/nu17020310>
- [21] Semerci, R., Akgün, M., & Kaya, N. (2025). Cultural and individual factors in AI-based nutrition recommendation systems: A literature review. *International Journal of Medical Informatics*, 185, 105–118. <https://doi.org/10.1016/j.ijmedinf.2025.105118>
- [22] Kok, W. E., Vinke, P. C., Schuurmans, M. J., & de Leeuw, J. R. J. (2022). The association between polypharmacy and malnutrition (risk) in older people: A systematic review. *Clinical Nutrition ESPEN*, 49, 163–172. <https://doi.org/10.1016/j.clnesp.2022.03.018>
- [23] Jyrkkä, J., Enlund, H., Lavikainen, P., Sulkava, R., & Hartikainen, S. (2011). Association of polypharmacy with nutritional status, functional ability and cognitive capacity over a three-year

period in an elderly population. *Pharmacoepidemiology and Drug Safety*, 20(5), 514–522. <https://doi.org/10.1002/pds.2116> PubMed

[24] Zanetti, M., Granzotto, M., Gortan Cappellari, G., & Barazzoni, R. (2023). Polypharmacy and malnutrition in older people: A narrative review. *Clinical Nutrition*, 42(9), 1688–1698. <https://doi.org/10.1016/j.clnu.2023.06.019>

[25] Fjeldstad, M., & Ranhoff, A. H. (2024). Malnutrition and polypharmacy in older adult patients receiving home care nursing services: A cross-sectional study. *Journal of the American Medical Directors Association*, 25(3), 401–407. <https://doi.org/10.1016/j.jamda.2023.10.022>